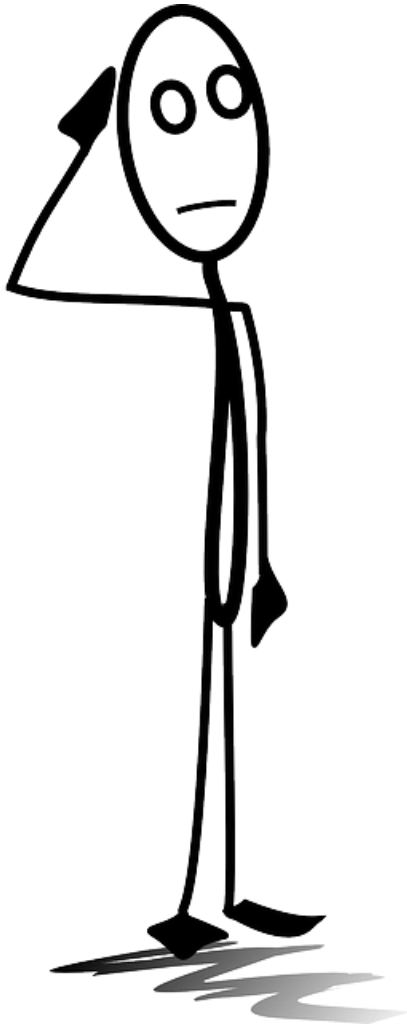
PHYSICS I
WEEK 6



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WEET 6: OUTLINE

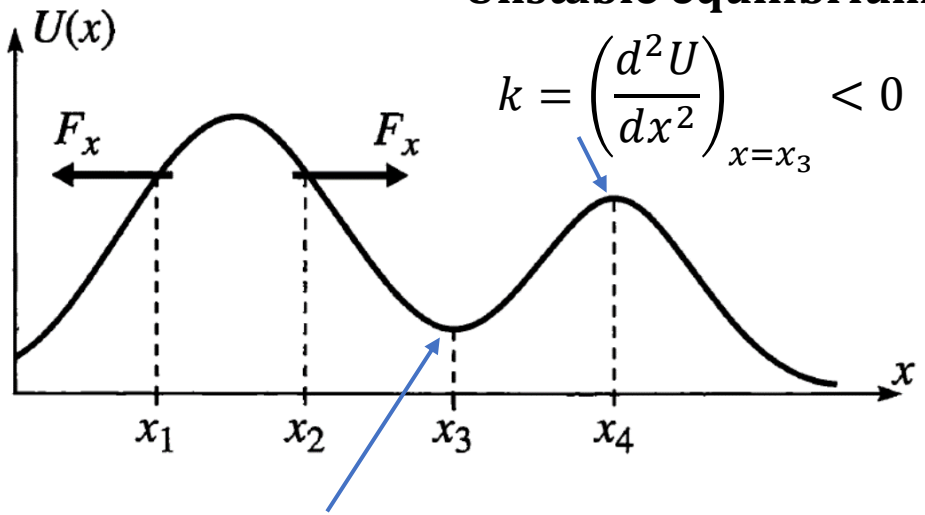
- Phase space (Thornton-Marion 3.4, 3.5)
- Bounded/Unbounded motion (Thornton-Marion 4.3, 4.4)



LINEAR 1D SYSTEMS: POTENTIAL ENERGY

$$U(x) = -\int_{x_0}^{x} F_x(x')\,dx' \quad \Rightarrow \quad F_x = -\frac{dU}{dx}$$

Unstable equilibrium



$$k = \left(\frac{d^2U}{dx^2}\right)_{x=x_3} < 0$$

Stable equilibrium (due to the restoring force)

$$k = \left(\frac{d^2U}{dx^2}\right)_{x=x_3} > 0 \qquad \omega = \sqrt{\frac{k}{m}}$$

Figure 4.10 The graph of potential energy $U(x)$ against x for any one-dimensional system can be thought of as a picture of a roller coaster track. The force $F_x = -dU/dx$ tends to push the object "downhill" as at x_1 and x_2. At the points x_3 and x_4, where $U(x)$ is minimum or maximum, $dU/dx = 0$ and the force is zero; such points are therefore points of equilibrium.

LINEAR 1D SYSTEMS: EXAMPLE

Linear harmonic oscillator $U(x) = Kx^2/2$ with $K > 0$

For any $E > 0$

$x_- \leq x \leq x_+$, where $x_\pm = \pm\sqrt{2E/K}$. ***Turning points***

Each phase trajectory is an ellipse given by

$$\frac{1}{2}m\dot{x}^2 + U(x) = E \quad \Rightarrow \quad \boxed{\frac{x^2}{(2E/K)} + \frac{\dot{x}^2}{(2E/m)} = 1} \quad \Rightarrow \quad \boxed{\frac{d\dot{x}}{dx} = -\omega_0^2 \frac{x}{\dot{x}}}$$

First-order differential equation

LINEAR 1D SYSTEMS: EXAMPLE

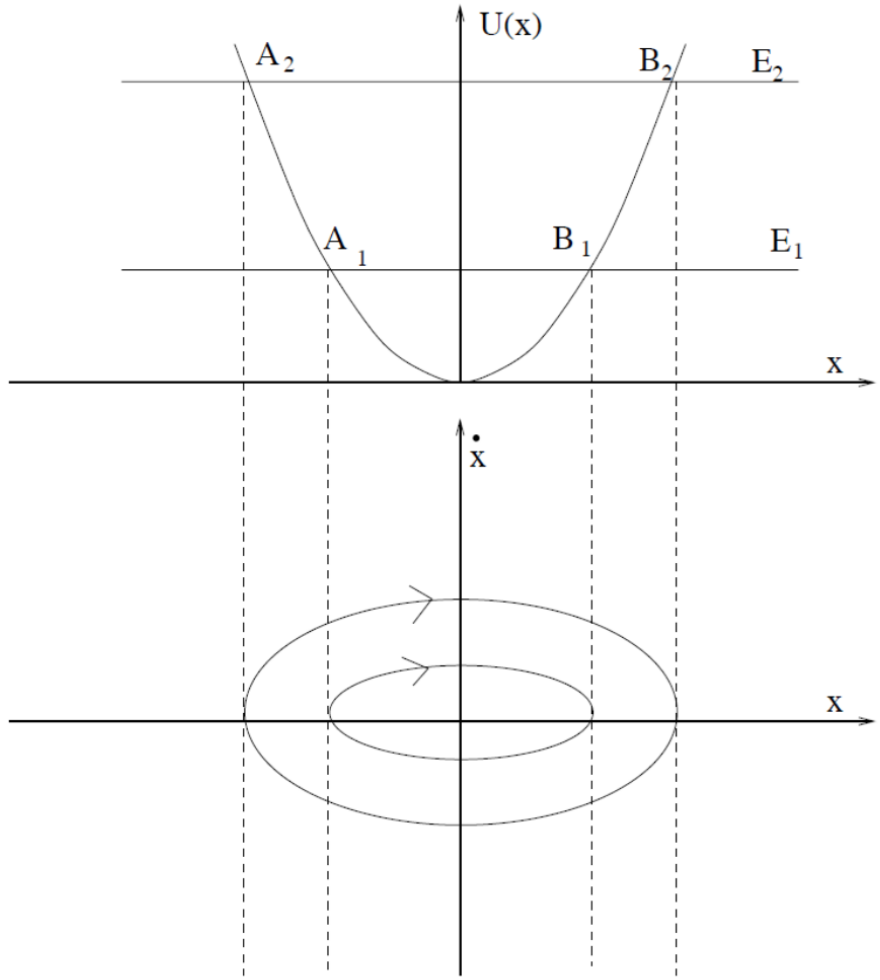
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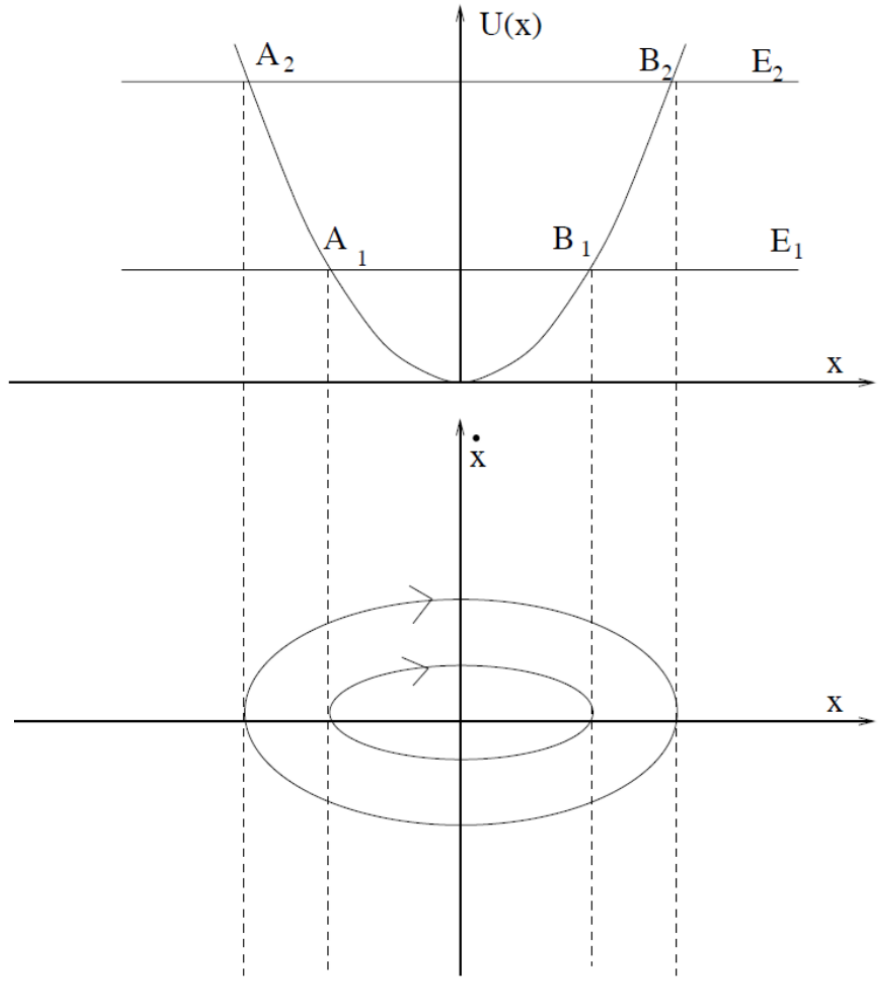
- Every E > 0 specifies a particular ellipse. The phase portrait is a family of such ellipses in the phase plane.
- The initial conditions determine the particular ellipse on which the system lies, as well as the precise point on it at which the motion starts.
- The sense or direction in which the ellipse is traversed as time progresses is not arbitrary – it is clockwise, as indicated by the arrows.

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Each phase trajectory is an ellipse given by

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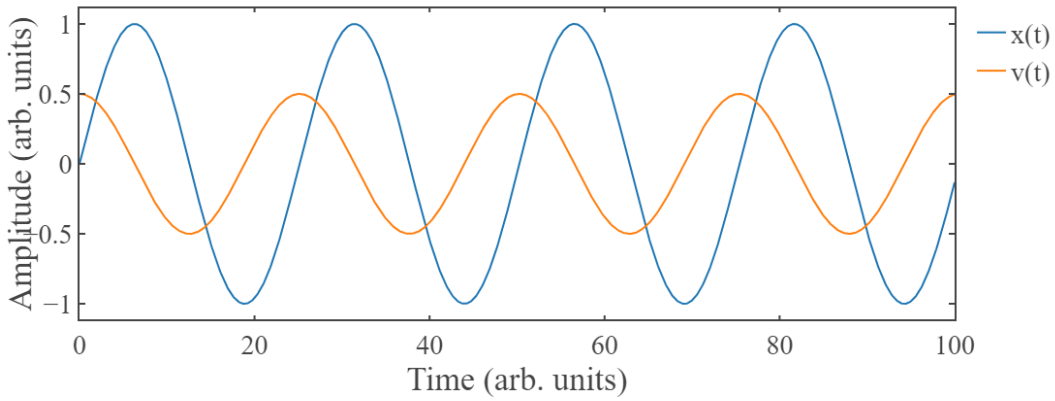


$$\ddot{x} = -\frac{k}{m}x = -\omega^2 x$$

Initial conditions

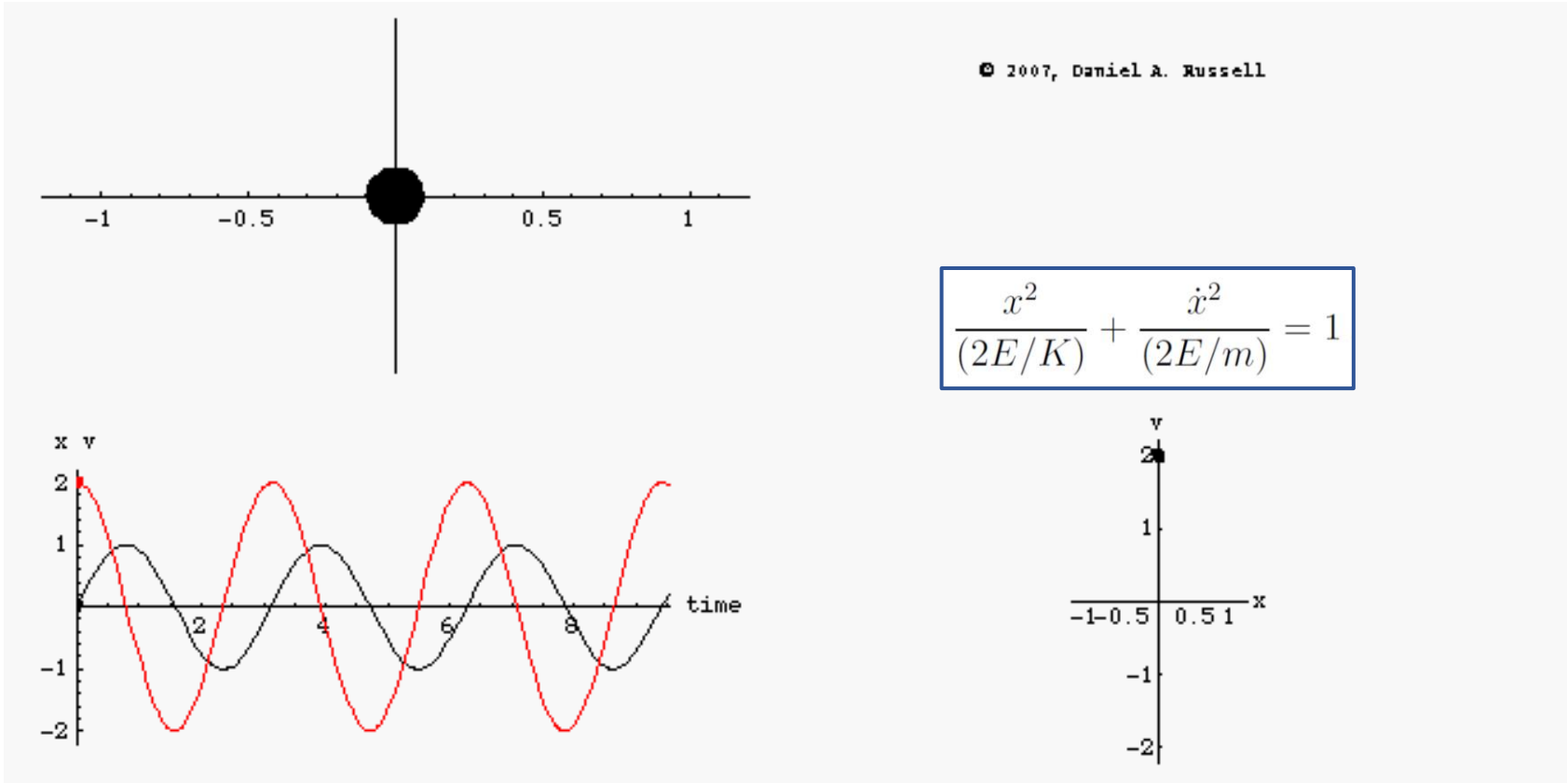
$x = 0$ **&** $v = v_0$

$$x(t) = \frac{v_0}{\omega}\sin(\omega t)$$



LINEAR 1D SYSTEMS: EXAMPLE

Linear harmonic oscillator $U(x) = Kx^2/2$ with $K > 0$



$$\frac{x^2}{(2E/K)} + \frac{\dot{x}^2}{(2E/m)} = 1$$

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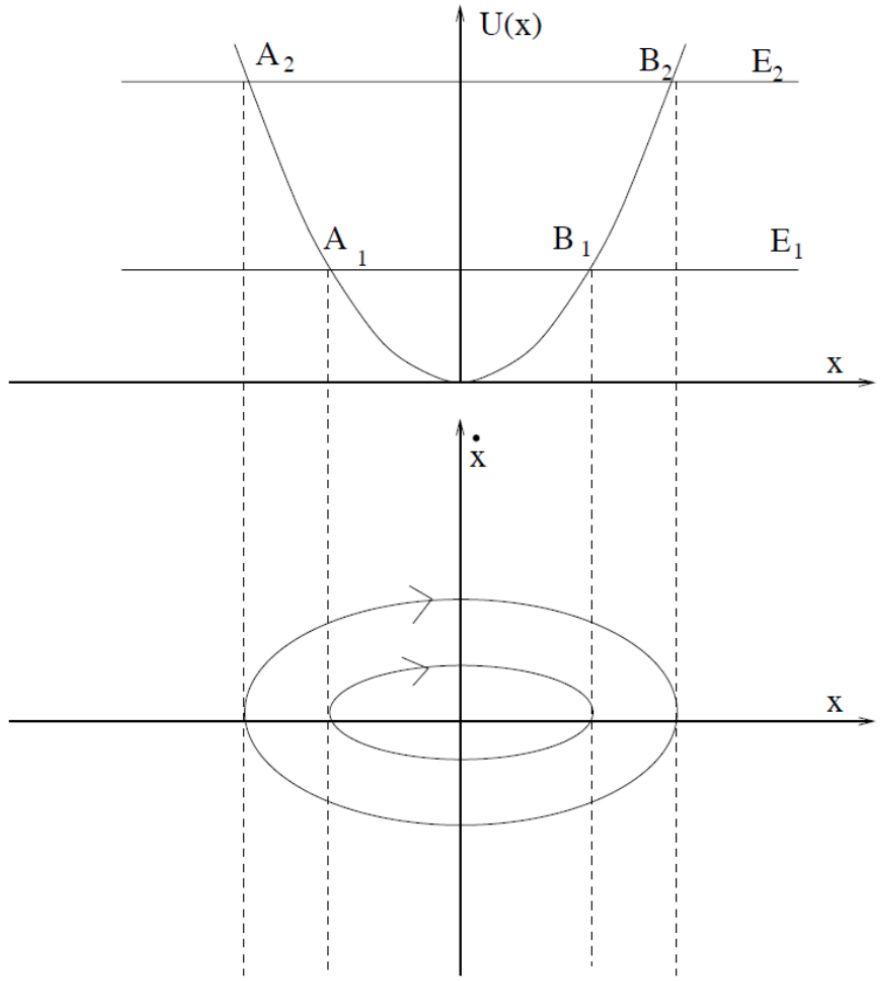
For any $E > 0$

$x_- \leq x \leq x_+$, where $x_\pm = \pm\sqrt{2E/K}$. ***Turning points***

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$$t = \int_{x_0}^{x} \frac{dx'}{\dot{x}(x')} = \sqrt{\frac{m}{2}} \int_{x_0}^{x} \frac{dx'}{\sqrt{E - U(x')}}$$



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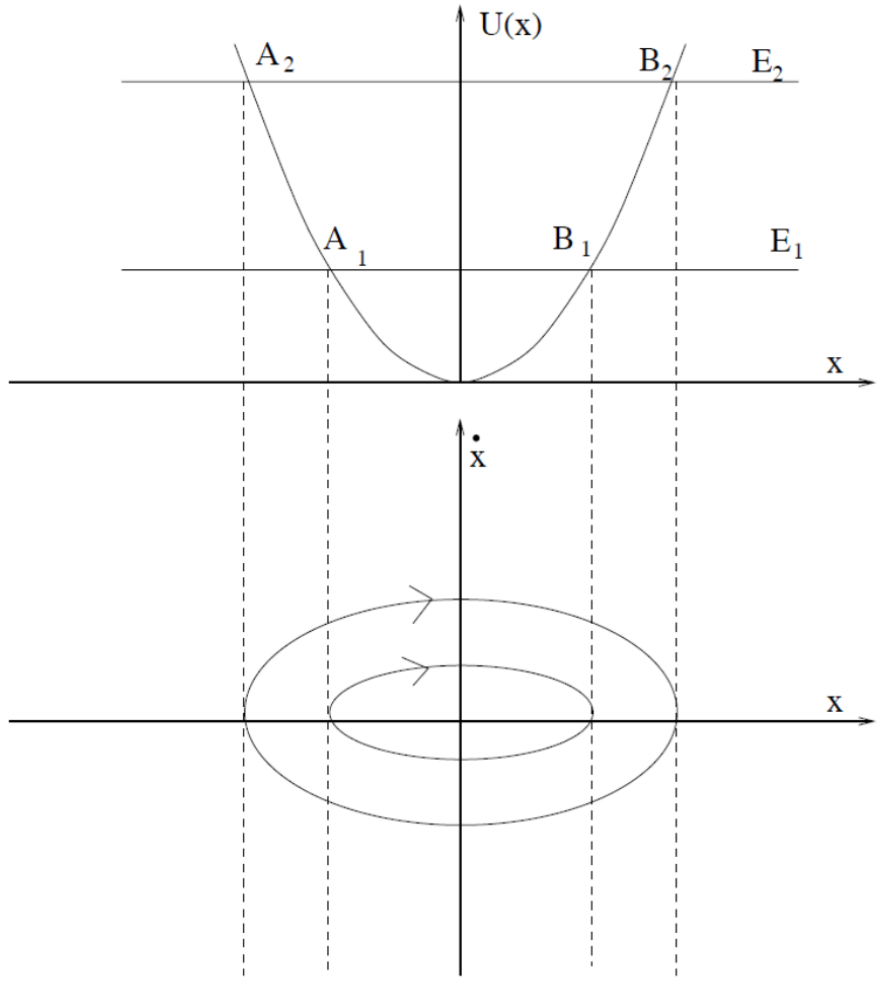
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$$t = \int_{x_0}^{x} \frac{dx'}{\dot{x}(x')} = \sqrt{\frac{m}{2}} \int_{x_0}^{x} \frac{dx'}{\sqrt{E - U(x')}}$$

Does not depend on energy!

$$T = 2\int_{A_1}^{B_1} \frac{\sqrt{m}dx}{\sqrt{(2E - Kx^2)}} \quad \Rightarrow \quad \boxed{T = 2\pi(m/K)^{1/2}}$$



LINEAR 1D SYSTEMS: EXAMPLE

Linear anharmonic oscillator

$$U(x) = (1/4)\lambda\, x^4$$

$$F(x) = -dU/dx = -\lambda\, x^3$$ ***Non-linear function***

$$(1/2)m\,\dot{x}^2 + (1/4)\lambda\, x^4 = E$$

LINEAR 1D SYSTEMS: EXAMPLE

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- The phase trajectories are no longer ellipses, but ovals!
- The explicit solution for $x(t)$ cannot be written in terms of elementary periodic functions like sines or cosines.

LINEAR 1D SYSTEMS: EXAMPLE

Linear anharmonic oscillator

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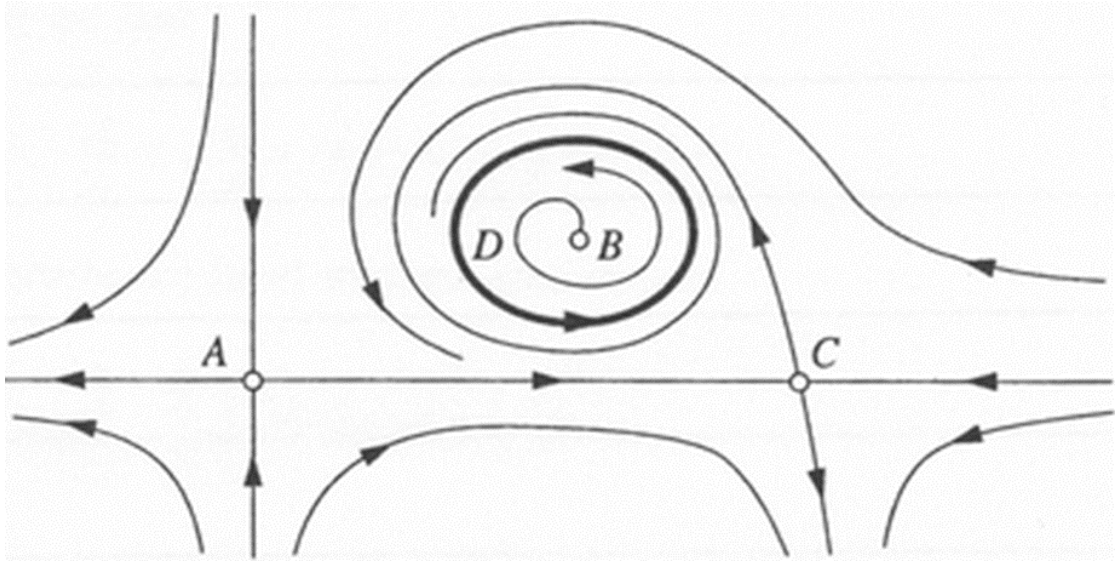
- The phase trajectories are no longer ellipses, but ovals!
- The explicit solution for $x(t)$ cannot be written in terms of elementary periodic functions like sines or cosines.
- Nonetheless, the time period corresponding to motion with any total energy E > 0 is easily found.

$$\text{turning points are at } x_\pm = \pm(4E/\lambda)^{1/4}$$

$$T = 2\int_{x_-}^{x_+} \frac{\sqrt{m}dx}{\sqrt{2(E-\lambda x^4/4)}} = 4\sqrt{\frac{2m}{\lambda}}\int_0^{x_+} \frac{dx}{\sqrt{x_+^4 - x^4}} = 4\left(\frac{m^2}{\lambda E}\right)^{1/4}\int_0^1 \frac{dx'}{\sqrt{1-x'^4}}$$ ***Depends on energy!***

- Except for simple harmonic motion, the time period for every other kind of periodic motion depends on the total energy (or, equivalently, on the amplitude of oscillation).

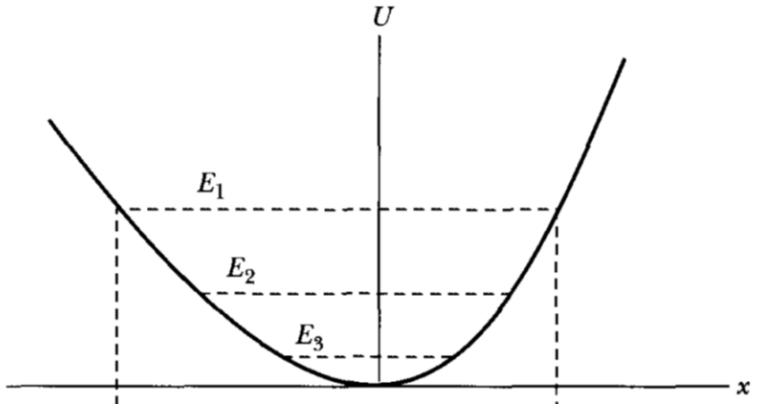
PHASE TRAJECTORIES: PROPERTIES



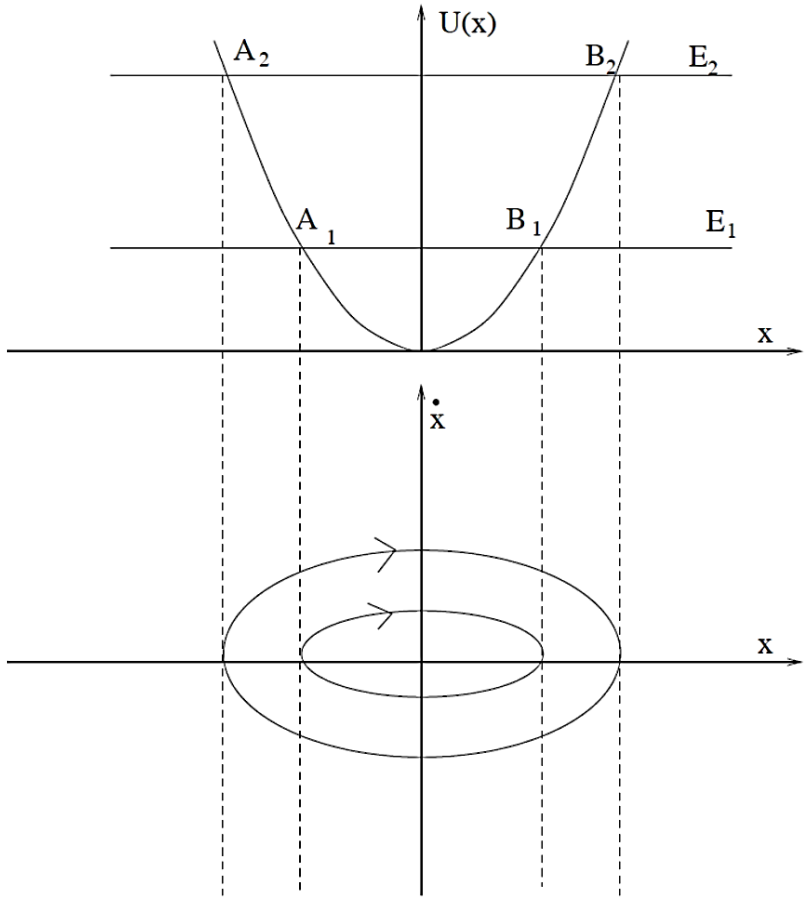
- The fixed points (A, B, and C) satisfy $\mathbf{F}(x) = 0$, and correspond to steady states or equilibria of the system.
- The closed orbits, (D) correspond to periodic solutions, i.e., solutions for which $x(T + t) = x(t)$ for all t, for some $T > 0$.
- The fixed points A, B, and C are unstable, because nearby trajectories tend to move away from them.

BOUNDED & UNBOUNDED MOTION

Example: Asymmetric potential

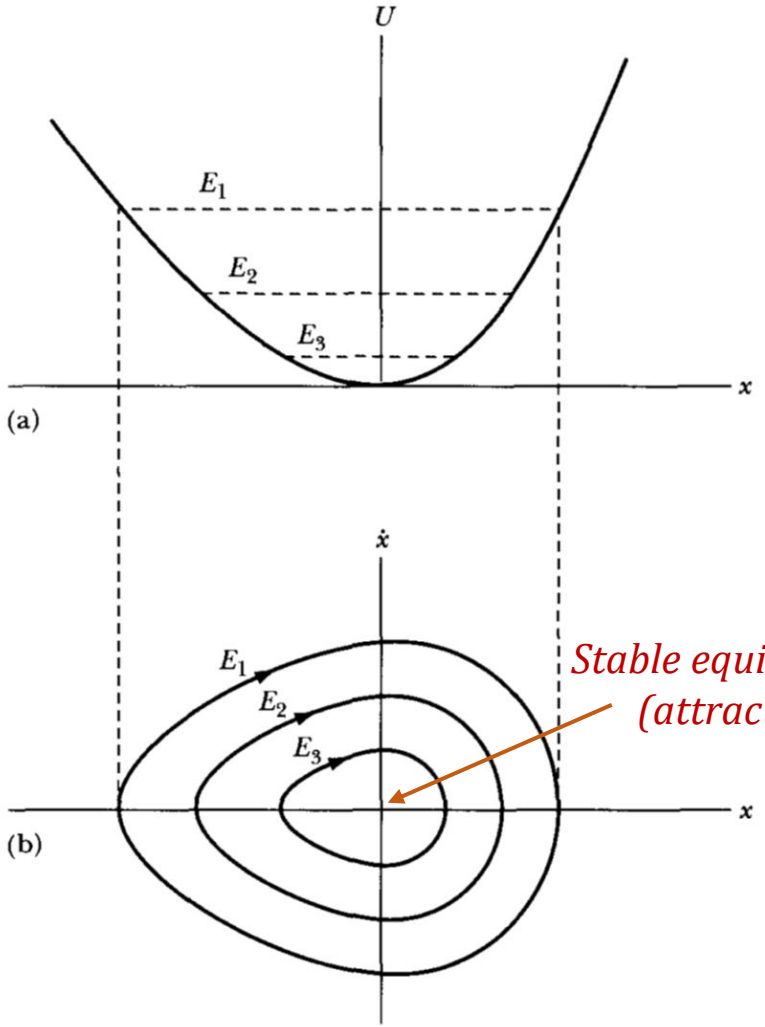


Example: Symmetric potential



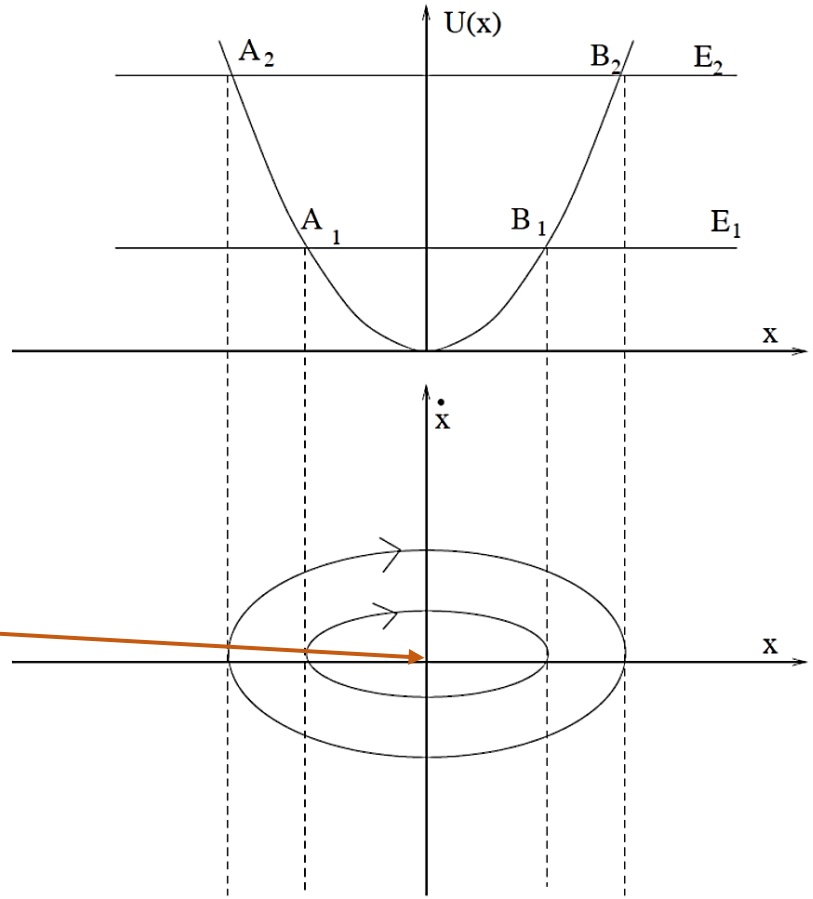
BOUNDED & UNBOUNDED MOTION

Example: Asymmetric potential



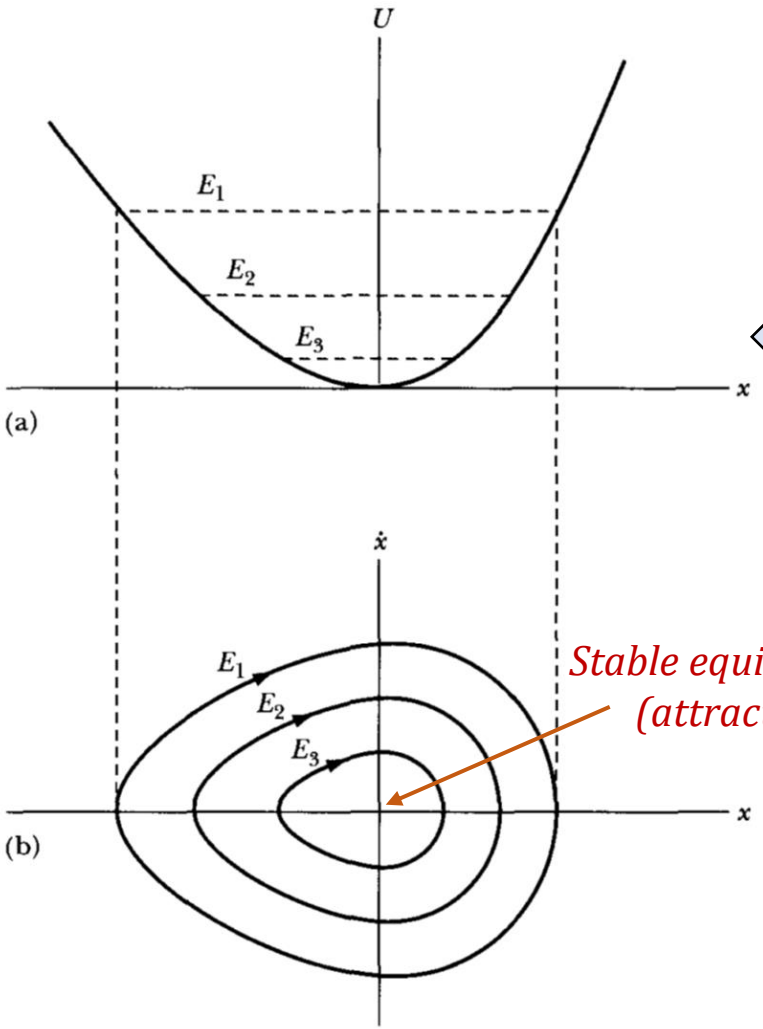
Stable equilibrium (attractor)

Example: Symmetric potential



BOUNDED & UNBOUNDED MOTION

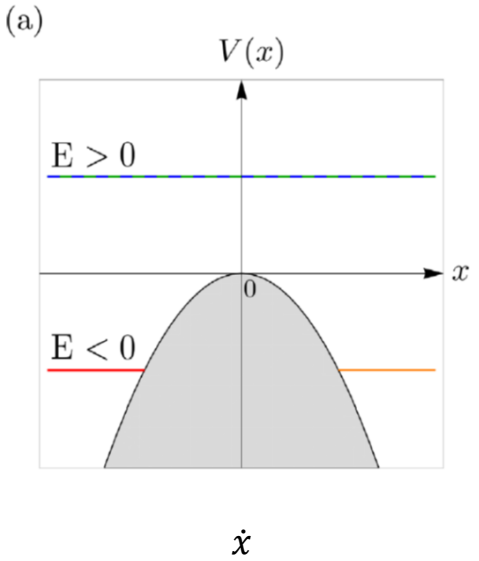
Example: Asymmetric potential



Stable equilibrium (attractor)

Bounded & Unbounded

Example: Inverted symmetric

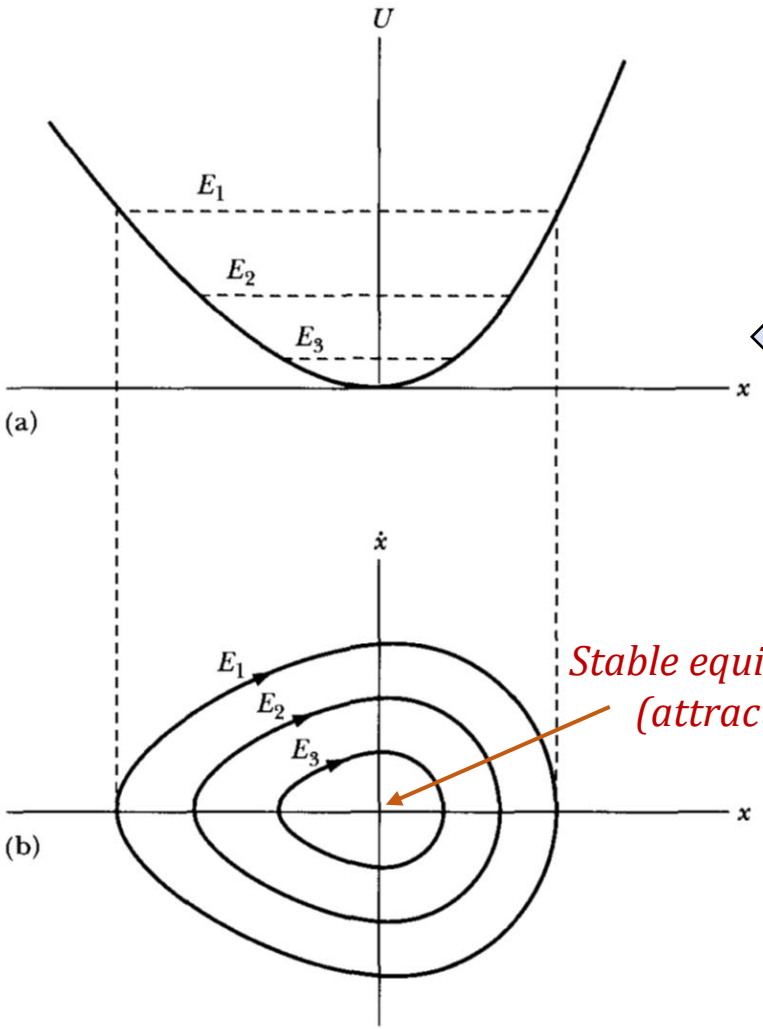


$\dot{x}$

AVS Quantum Sci.4, 024402 (2022)

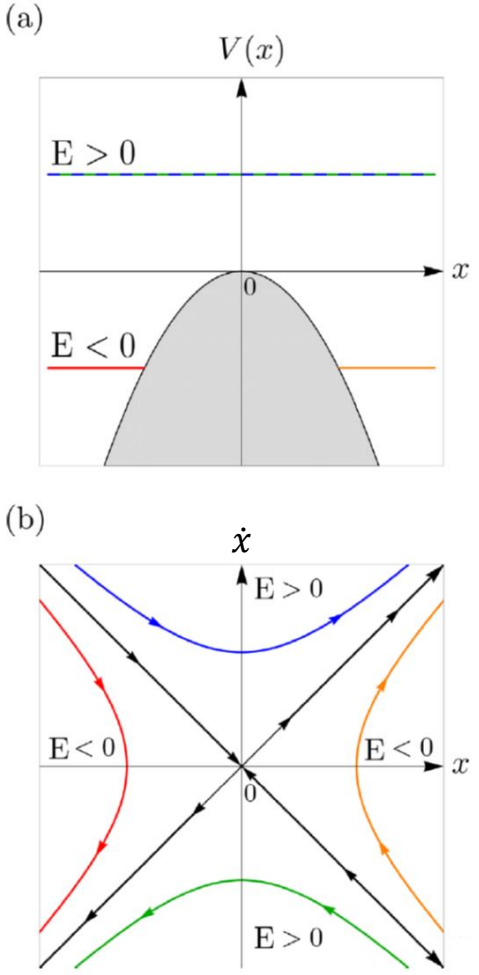
BOUNDED & UNBOUNDED MOTION

Example: Asymmetric potential



Bounded & Unbounded

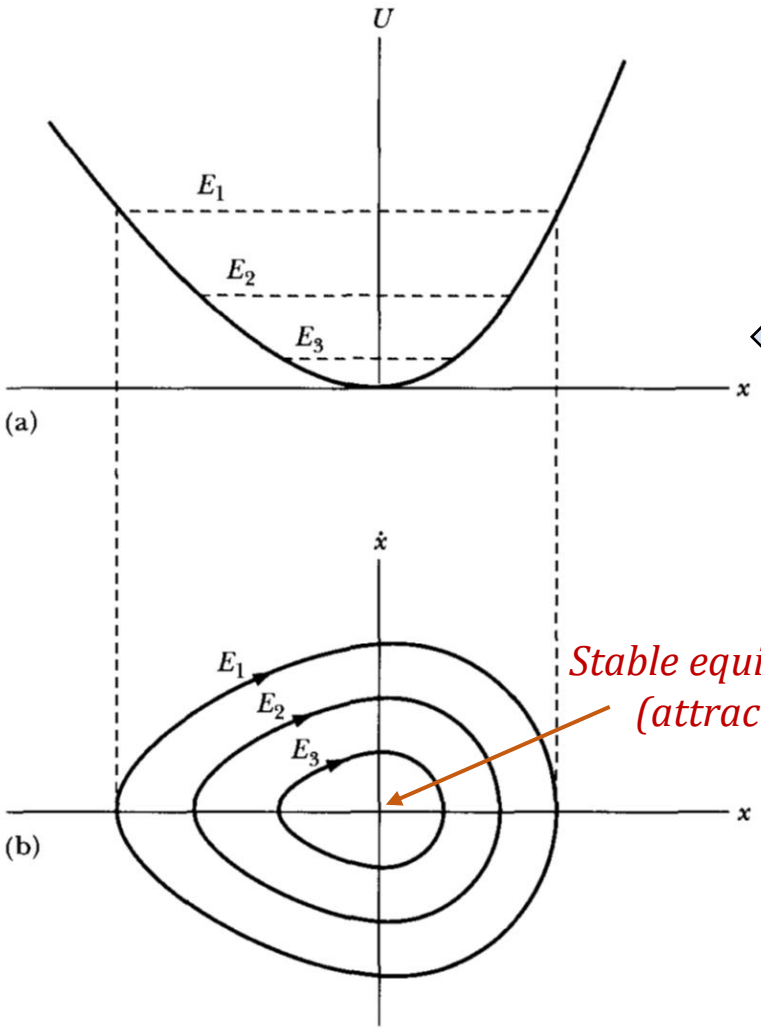
Example: Inverted symmetric



AVS Quantum Sci.4, 024402 (2022)

BOUNDED & UNBOUNDED MOTION

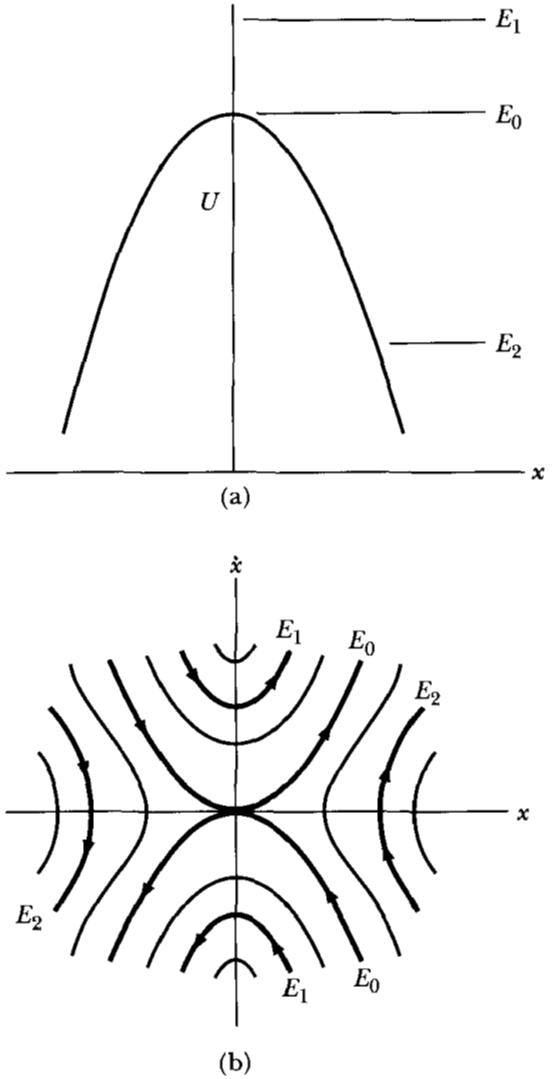
Example: Asymmetric potential



Stable equilibrium (attractor)

Bounded & Unbounded

Example: Inverted asymmetric



REVIEW: DAMPED OSCILLATIONS

Undamped β is zero

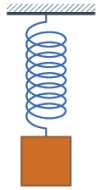
$$x(t) = C_1 e^{i\omega_o t} + C_2 e^{-i\omega_o t}$$

Weak Damping $\beta < \omega_o$

$$x(t) = A e^{-\beta t} \cos(\omega_1 t - \delta)$$

Strong Damping $\beta > \omega_o$

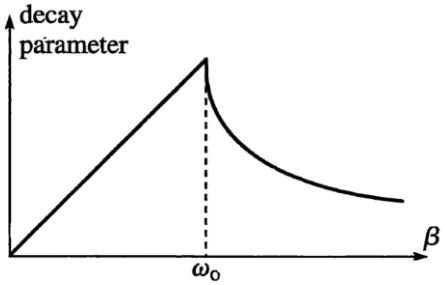
$$x(t) = C_1 e^{-\left(\beta - \sqrt{\beta^2 - \omega_o^2}\right)t} + C_2 e^{-\left(\beta + \sqrt{\beta^2 - \omega_o^2}\right)t}$$

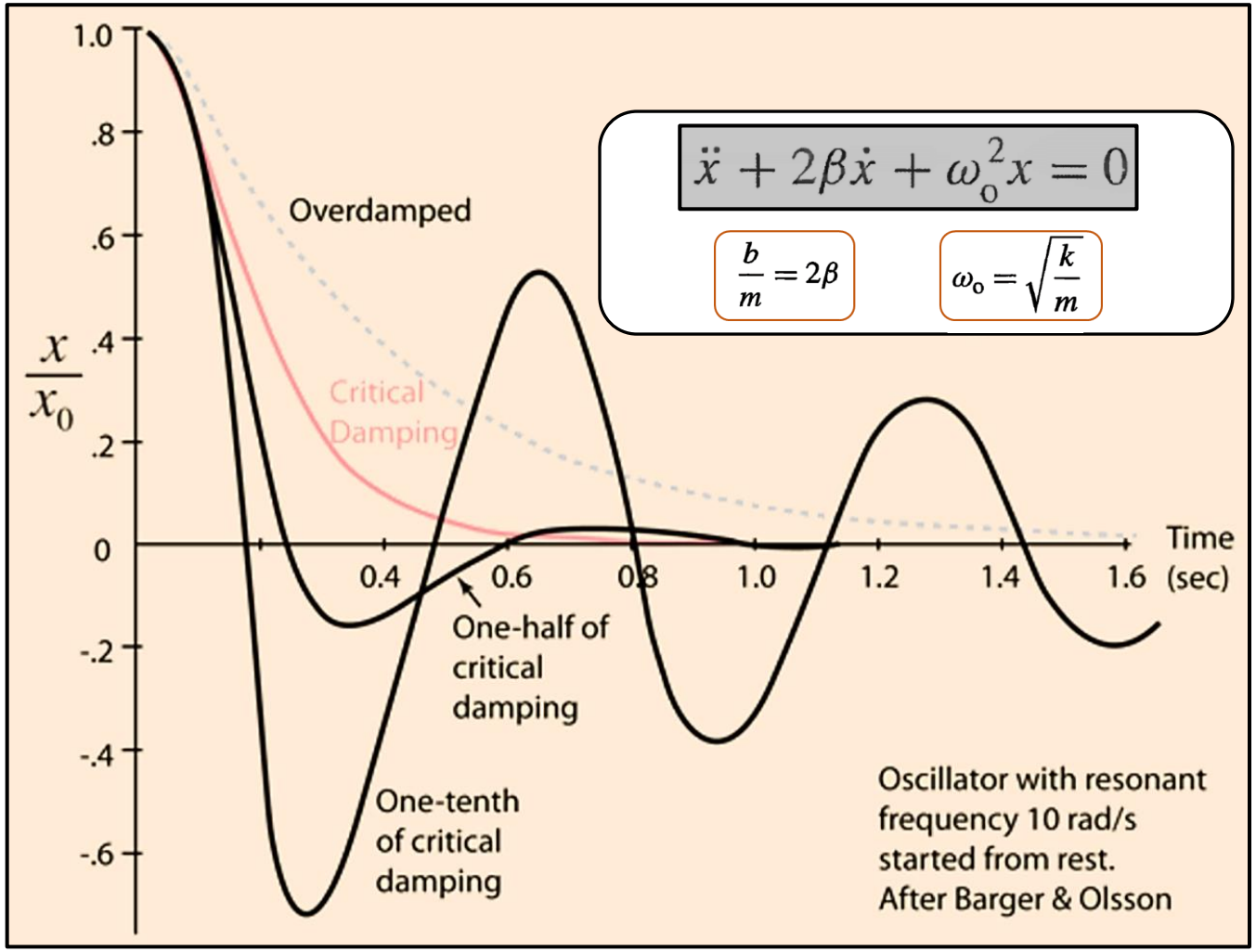


Critical Damping $\beta = \omega_o$

$$x(t) = C_1 e^{-\beta t} + C_2 t e^{-\beta t}$$

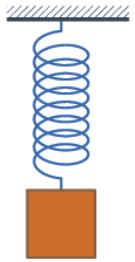
damping	β	decay parameter
none	$\beta = 0$	0
under	$\beta < \omega_o$	β
critical	$\beta = \omega_o$	β
over	$\beta > \omega_o$	$\beta - \sqrt{\beta^2 - \omega_o^2}$





LINEAR 1D SYSTEMS: EXAMPLE

Underdamped oscillator



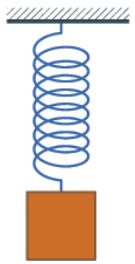
Weak Damping $\beta < \omega_o$

$$\boxed{x(t) = Ae^{-\beta t}\cos(\omega_1 t - \delta)}$$

$$\omega_1 = \sqrt{\omega_o^2 - \beta^2} = \text{frequency of damped oscillator}$$

LINEAR 1D SYSTEMS: EXAMPLE

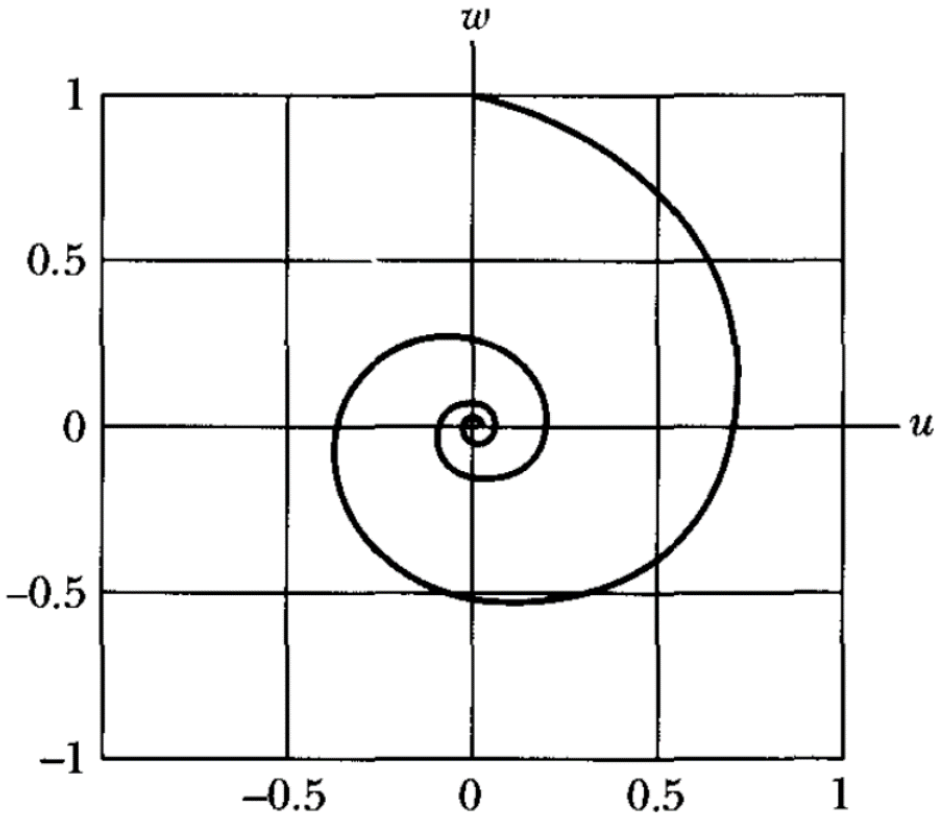
Underdamped oscillator



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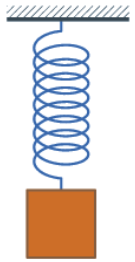
$$u = \omega_1 x, \quad w = \beta x + \dot{x}$$

$$u = \omega_1 A e^{-\beta t}\cos(\omega_1 t - \delta)$$

$$w = -\omega_1 A e^{-\beta t}\sin(\omega_1 t - \delta)$$

LINEAR 1D SYSTEMS: EXAMPLE

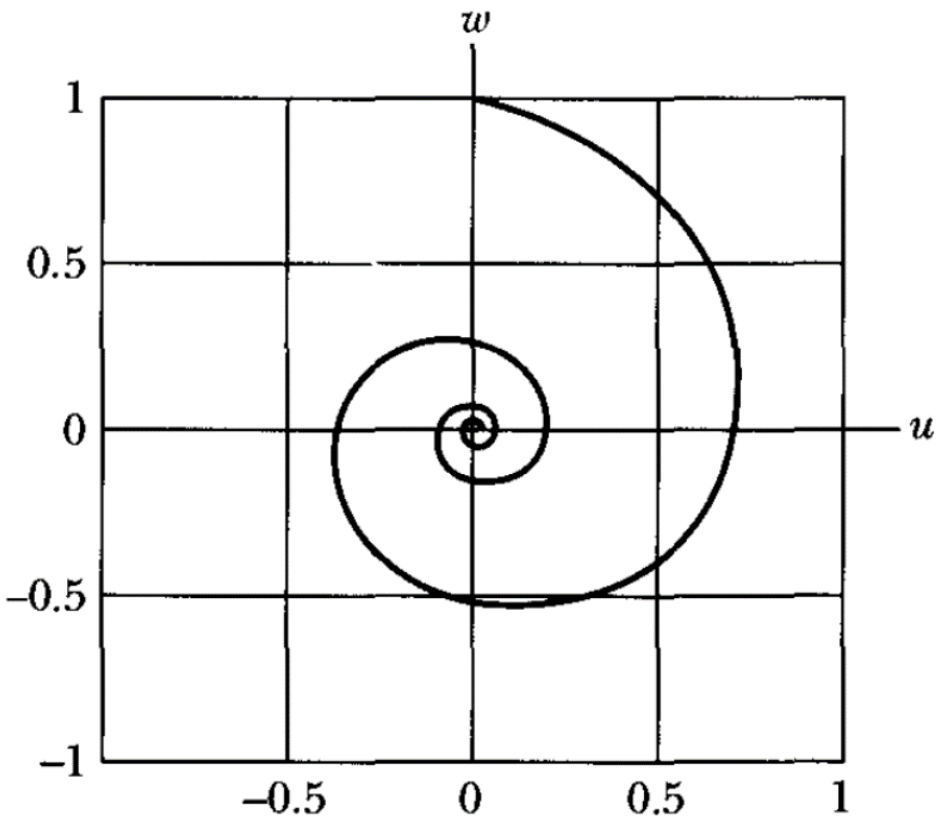
Underdamped oscillator

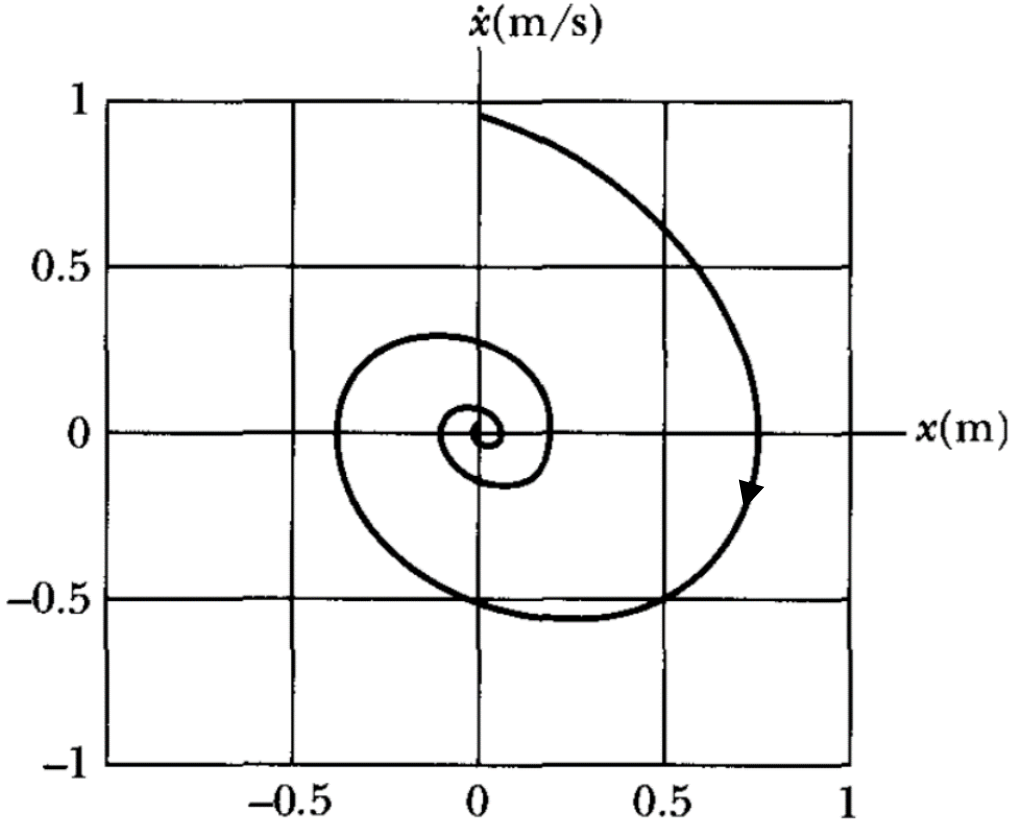


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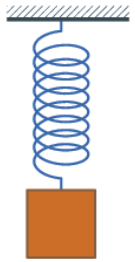
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LINEAR 1D SYSTEMS: EXAMPLE

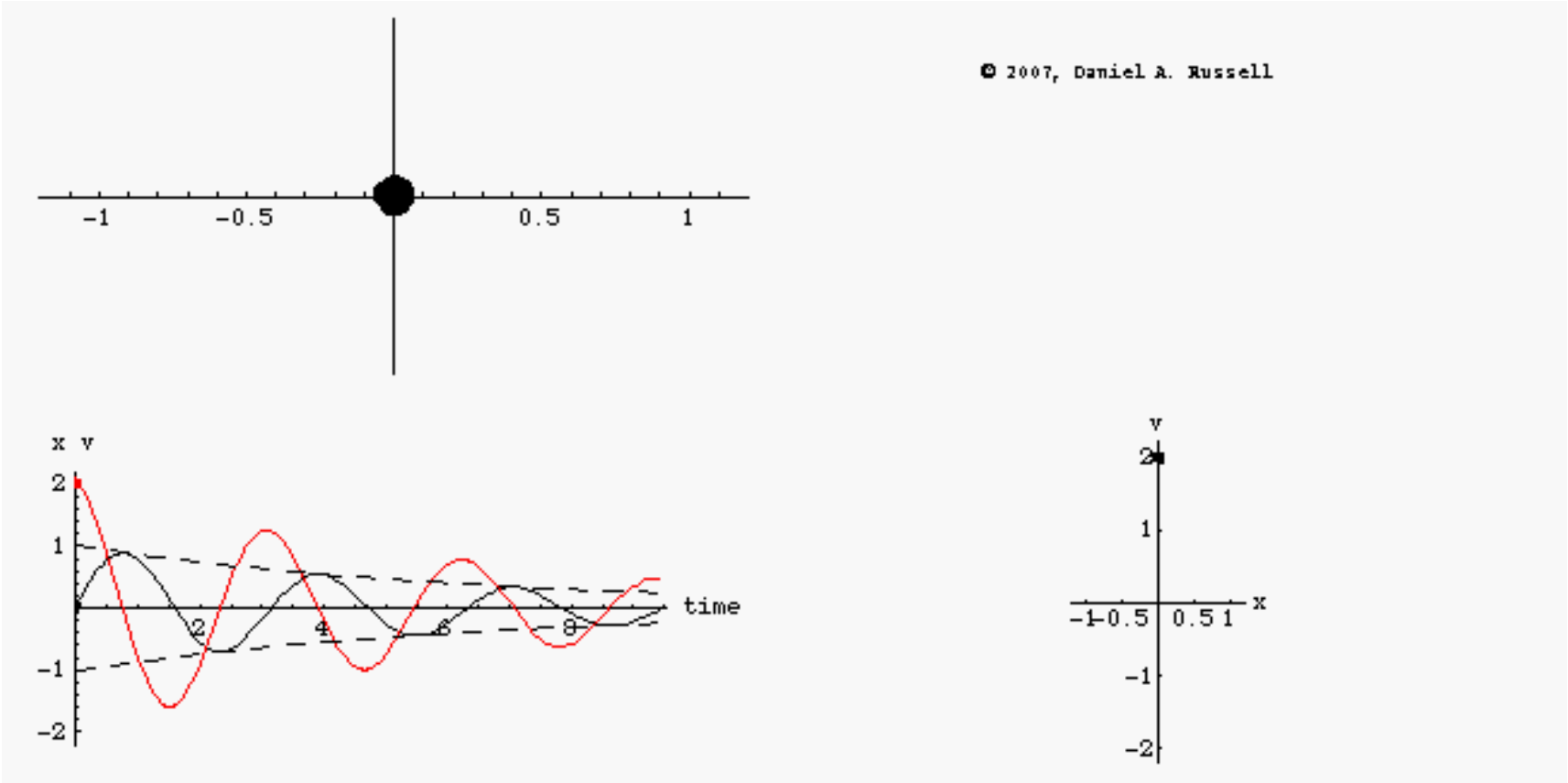
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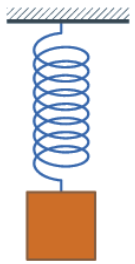
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